

MATH3026/4022: Time Series Analysis/Time Series and Forecasting
Exercises Sheet 2

To be covered during the examples class at 16:00 on Monday 23rd February. These exercises do not count toward your final module mark, however it is a good idea to attempt these questions prior to the examples class.

1. Suppose that $\{Z_t\}$ is a white noise process with variance σ_z^2 . For each of the following processes, $\{X_t\}$, answer the following: (a) Is the process stationary? (b) If the process is stationary, what is its mean? (c) If the process is stationary, what is its autocovariance function?

(i) $X_t = \frac{5}{4}X_{t-1} - \frac{3}{8}X_{t-2} + Z_t$.

(ii) $X_t = \frac{5}{4}X_{t-1} + \frac{1}{8}X_{t-2} - \frac{3}{8}X_{t-3} + Z_t$.

(iii) $X_t = \sum_{j=0}^{\infty} \psi_j Z_{t-j}$ where $\psi_j = (1+j)^{-1}$.

2. The process $\{X_t\}$ is a mean zero $AR(2)$ process such that

$$X_t - (\gamma_1 + \gamma_2)X_{t-1} + \gamma_1\gamma_2X_{t-2} = Z_t$$

where $|\gamma_1| < 1$, $|\gamma_2| < 1$ and $\{Z_t\}$ is a white noise process with variance σ_z^2 .

- (a) Explain why $\{X_t\}$ is a stationary process.
 (b) Show that $\{X_t\}$ can be written as an $MA(\infty)$ process, with the form

$$X_t = \frac{1}{\gamma_2 - \gamma_1} \sum_{k=0}^{\infty} (\gamma_2^{k+1} - \gamma_1^{k+1}) Z_{t-k}.$$

- (c) Derive the autocovariance function of $\{X_t\}$.

3. For each of the following ARMA processes, determine: (i) whether or not the process is stationary; (ii) whether or not the process is invertible. In each case, $\{Z_t\}$ is a white noise process with variance σ_z^2 .

(a) $X_t + 0.2X_{t-1} - 0.48X_{t-2} = Z_t$

(b) $X_t + 1.9X_{t-1} + 0.88X_{t-2} = Z_t + 0.2Z_{t-1} + 0.7Z_{t-2}$

(c) $X_t + 0.6X_{t-1} = Z_t + 1.2Z_{t-1}$

(d) $X_t + 1.8X_{t-1} + 0.81X_{t-2} = Z_t$

(e) $X_t + 1.6X_{t-1} = Z_t - 0.3Z_{t-1}$

4. Suppose that the time series $\{Y_t\}$ is defined

$$Y_t = X_t + W_t$$

where $\{W_t\}$ is a white noise process with variance σ_w^2 and $\{X_t\}$ is an $AR(1)$ process such that

$$X_t = \phi X_{t-1} + Z_t$$

where $|\phi| < 1$, $\{Z_t\}$ is a white noise process with $\text{Var}(Z_t) = \sigma_z^2$ and $\mathbb{E}(W_s Z_t) = 0$ for all s and t .

- (a) Show that the process $\{Y_t\}$ is stationary and determine its autocovariance function.
- (b) Show that the process $\{U_t\}$, where $U_t = Y_t - \phi Y_{t-1}$, is an $MA(1)$ process.
- (c) Show that $\{Y_t\}$ is an $ARMA(1,1)$ process and determine the parameters of this process in terms of ϕ , σ_w^2 and σ_z^2 .

5. The process $\{X_t\}$ is the $AR(2)$ process, written

$$X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + Z_t,$$

where $\{Z_t\}$ is a white noise process with variance σ_z^2 . Suppose that $\rho(1)$ and $\rho(2)$ denote the autocorrelations for this process at lags 1 and 2, respectively.

Prove that the following conditions are necessary and sufficient for the $AR(2)$ process to be weakly stationary:

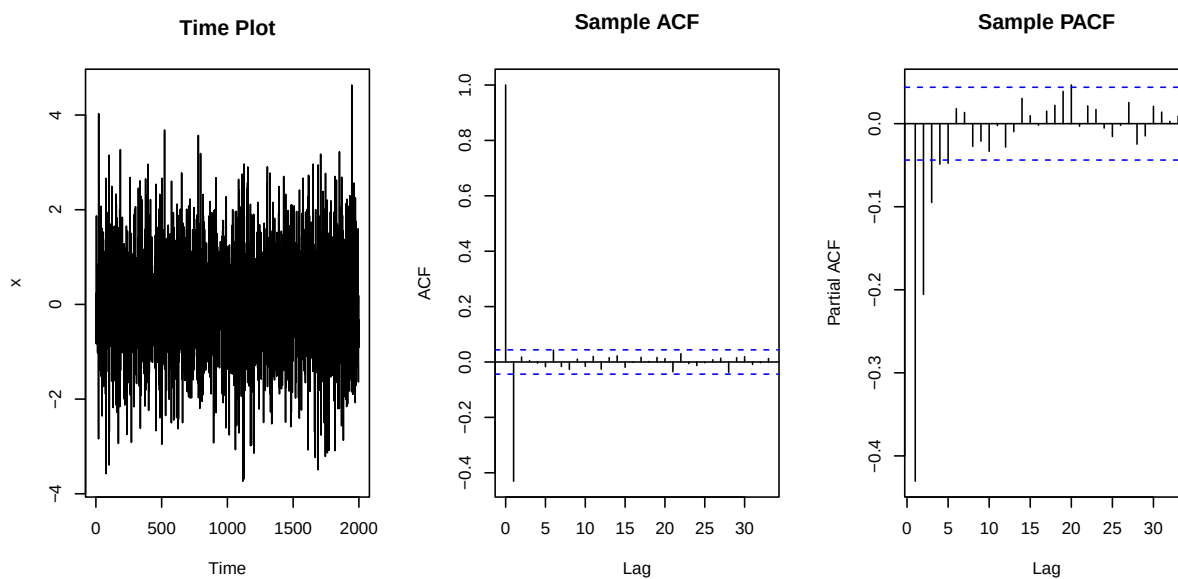
$$\begin{aligned} -1 &< \rho(1) < 1; \\ -1 &< \rho(2) < 1; \\ \rho(1)^2 &< \frac{1}{2}[\rho(2) + 1]. \end{aligned}$$

6. The process $\{X_t\}$ is an $AR(2)$ process where

$$X_t = \frac{2}{3}X_{t-1} - \frac{2}{9}X_{t-2} + Z_t.$$

- (a) Write down the Yule-Walker equations for this process.
- (b) By solving the Yule-Walker equations, determine the autocorrelation function of $\{X_t\}$.

7. A statistician would like to fit an ARMA model to a time series, where 2000 sequential observations have been observed. The observations provided sample autocorrelations and sample partial autocorrelations, shown in the plots below, together with a time plot.



- (a) Using these plots, identify the type of ARMA model that might fit the data. Justify your answer.
- (b) Determine possible parameter values for the process chosen in (a).